

Maziar Zamanpour

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SUMMARY

Ph.D. candidate in Transportation Engineering (graduating Jan 2027) with 3+ years of research experience in traffic operations, intelligent transportation systems, and transportation data analysis. Hands-on experience with field data collection, traffic simulation, and end-to-end technical report writing for MnDOT-funded projects. Background in analytical modeling, statistical analysis, and Python/R programming.

TECHNICAL SKILLS

- **Transportation & Traffic:** Traffic simulation (SUMO, VISSIM, Synchro), signal timing & SPaT data, connected vehicle (V2I) systems, highway design, field data collection & processing.
- **Programming & Data:** Python (NumPy, Pandas, scikit-learn), Java, R, SQL, Data Visualization (Tableau).
- **Analytics:** Predictive Modeling (Regression, KNN, Decision Trees), Time Series / Kalman Filter, Cost-Benefit Analysis, Statistics.
- **Software & Tools:** Civil 3D, AutoCAD, Bluebeam, SAP, Git/GitHub, Microsoft Office, Google Workspace.
- **Technical Writing:** End-to-end report authoring for state DOT projects, literature reviews, methodology documentation, research presentations (TRB, INFORMS, ITS Minnesota).

EXPERIENCE

Transportation Research Scientist

University of Minnesota - Minneapolis, MN

August 2022 - present

- Developed a **real-time red light running warning system** using connected V2I technology and model-predictive control; led **field data collection** and public-road testing in Scott County, MN for MnDOT (Demo).
- Authored **two technical reports** for the Minnesota Department of Transportation covering system design, field test results, and deployment recommendations.
- Built an optimization model for **pedestrian walkway network design** across transportation networks using Benders Decomposition (Python, Gurobi); reduced computation time by 90% for large urban networks (GitHub).
- Conducted **cost-benefit analysis** of connected eco-speed control deployment using SPaT data; demonstrated payback within 3 years, informing MnDOT corridor deployment strategy.
- Applied gradient descent optimization and trajectory prediction to develop an individualized eco-speed advisory model with **13–15% energy savings**; validated on real-world testbed.
- Extended a **dynamic network loading simulation** (Java) to capture energy-efficient autonomous vehicle behavior and route choice interactions in mixed traffic (GitHub).

Optimization & Product Development — Entrepreneurial Lead

National Science Foundation I-Corps Program

May 2024 - present

- Built dispatch optimization using integer programming (CPLEX) and ML-based demand forecasting for a car-sharing platform; modeled **75% cost reduction** vs. personal vehicle ownership.
- Secured **\$50K grant**; led 100+ customer discovery interviews to validate product-market fit.

Statistics and Probability — Course Instructor

University of Minnesota

August 2023 - present

- Lectured in discussion sessions, designed and presented course materials and problems.
- Covered probability, causal inference, Bayesian methods, distributions, point estimation, regression.

EDUCATION

University of Minnesota - Minneapolis, MN

Dec 2023 – Jan 2027

Ph.D., Civil Engineering: Transportation (+ Industrial Engineering Minor)

GPA 3.7

Coursework: *Transportation Data Analysis, Highway Design & Traffic Operations, Transport Cyber-Physical Systems, Decision Analysis*

University of Minnesota - Minneapolis, MN

Aug 2022 – Dec 2023

M.S., Civil Engineering: Transportation

GPA 3.6

University of Tehran - Tehran

Sep 2018 – May 2022

B.S., Civil Engineering

GPA 3.7

SELECTED PROJECTS

DoorDash Delivery Optimization

Fall 2025

- Formulated driver-delivery assignment as a MIP (PuLP, Gurobi); designed greedy heuristic for real-time dynamic assignment.

Airbnb Price Prediction

Fall 2023

- Built predictive pipeline: data cleaning, feature engineering, modeling (regression, trees, KNN) in R & Python; visualized in Tableau.

PUBLICATIONS & AWARDS

- **7 publications** (3 peer-reviewed journals incl. Transportation Research Part C; 2 MnDOT technical reports; 2 under review). Presented at INFORMS, TRB, ITS Minnesota.
- Won **Matthew J. Huber Award** for excellence in transportation research (2024). Won two **TRB Travel Awards** (2023, 2025).